

Autonomous Obstacle-Avoidance Robot Car Using Arduino and Ultrasonic Sensing

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An autonomous obstacle-avoidance robot car was designed, constructed, and tested using an Arduino microcontroller, an HC-SR04 ultrasonic distance sensor, an L298N motor driver module, DC motors, LEDs, and a piezoelectric buzzer. The system was programmed to drive forward at full speed when no obstacle was detected, reduce speed and emit an intermittent buzzer warning when an obstacle was detected within 20 cm, and halt completely with a red LED indicator and continuous buzzer alert when the obstacle was within 10 cm. The robot successfully demonstrated autonomous obstacle avoidance at distances of 1–1.5 m from walls. Limitations of the HC-SR04 sensor, including narrow beam angle and specular reflection effects, were identified and discussed.

I. INTRODUCTION

Autonomous mobile robots that can detect and avoid obstacles are fundamental systems in robotics and embedded electronics. This project integrated digital and analog electronics concepts — including PWM motor speed control, digital sensor interfacing, and conditional logic — into a single embedded system built around the Arduino Uno microcontroller. The goal was to produce a robot car that could navigate toward a wall and automatically slow down and stop before collision, mimicking the behavior of automotive parking sensors. A push-button was also incorporated so that the motor would not engage unless a valid sensor reading was detected, adding a layer of safety logic to the system.

II. THEORY

A. Ultrasonic Distance Sensing

The HC-SR04 ultrasonic sensor measures distance by emitting a burst of 40 kHz ultrasonic pulses from its transmitter (TRIG pin) and measuring the time for the echo to return to the receiver (ECHO pin). The distance to the obstacle is calculated as:

$$d = \frac{v_s \cdot t}{2} \quad (1)$$

where $v_s \approx 343$ m/s is the speed of sound in air at room temperature and t is the round-trip travel time of the pulse. The factor of 2 accounts for the pulse traveling to the obstacle and back. In Arduino code this simplifies to:

$$d [\text{cm}] = \frac{t [\mu\text{s}]}{58} \quad (2)$$

The HC-SR04 has a rated sensing range of 2–400 cm and an effective beam angle of approximately 15°. Targets that are outside this cone, at highly oblique angles,

or in large open spaces may reflect the ultrasonic pulse away from the receiver, causing missed or invalid readings.

B. PWM Motor Speed Control

The DC motors are driven by the L298N H-bridge motor driver, which accepts a Pulse Width Modulation (PWM) signal from the Arduino. The average voltage delivered to the motor is proportional to the duty cycle D :

$$\bar{V}_{motor} = D \cdot V_{supply} \quad (3)$$

where $D \in [0, 1]$ and V_{supply} is the motor supply voltage. In Arduino, the duty cycle is controlled by the `analogWrite()` function, which accepts an integer from 0 (motor off) to 255 (full speed), allowing continuous speed control.

C. Conditional Control Logic

The robot's behavior is governed by a three-zone proximity system based on distance thresholds:

$$\text{State} = \begin{cases} \text{Full speed, green LED} & d > 20 \text{ cm} \\ \text{Slow + yellow LED + beep} & 10 < d \leq 20 \text{ cm} \\ \text{Stop + red LED + alarm} & d \leq 10 \text{ cm} \end{cases} \quad (4)$$

III. APPARATUS

The following components were used:

- Arduino Uno microcontroller
- HC-SR04 ultrasonic distance sensor
- L298N dual H-bridge motor driver module

- 2× DC motors (mounted on robot chassis)
- 2WD robot car chassis kit with wheels
- Red, yellow, and green LEDs with 220 Ω resistors
- Piezoelectric buzzer
- Push-button with 10 k Ω pull-up resistor
- 9 V battery (motor power via L298N)
- USB power bank (Arduino power)
- Solderless breadboard and jumper wires

IV. CIRCUIT DESIGN AND PROCEDURE

A. Circuit Construction

The HC-SR04 TRIG and ECHO pins were connected to Arduino digital pins 9 and 10 respectively. The three LEDs were connected through 220 Ω current-limiting resistors to digital pins 4 (red), 5 (yellow), and 6 (green), with cathodes tied to ground. The buzzer was connected between digital pin 7 and ground. A push-button was wired between digital pin 12 and ground with a 10 k Ω pull-up resistor, so that the motor would only engage when the button was pressed and a valid sensor reading was available. The L298N motor driver IN1–IN4 pins were connected to Arduino digital pins 2, 3, 8, and 11, and the PWM enable pins (ENA, ENB) were connected to Arduino PWM-capable pins for speed control. The L298N was powered by a dedicated 9 V battery, and the Arduino was powered separately via a USB power bank to prevent motor current surges from resetting the microcontroller.

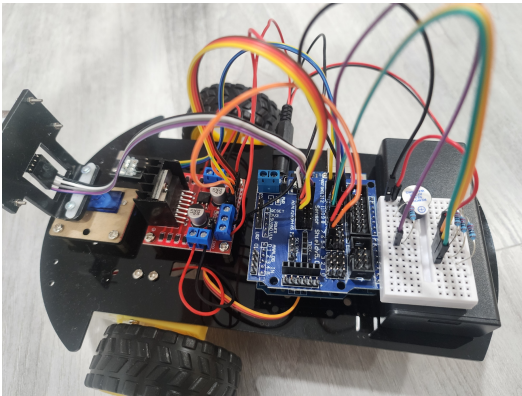


FIG. 1. Breadboard circuit layout showing the Arduino Uno, HC-SR04 ultrasonic sensor, L298N motor driver, LEDs, buzzer, and push-button connections.

B. Chassis Assembly

The breadboard circuit was mounted on a 2WD robot car chassis with the two DC motors connected to the L298N output terminals. The HC-SR04 ultrasonic sensor was mounted at the front of the chassis facing forward, and the LED indicators and buzzer were positioned for visibility during testing.

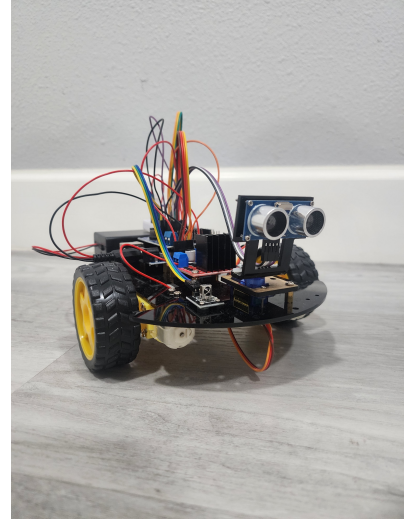


FIG. 2. Assembled obstacle-avoidance robot car with the Arduino circuit mounted on the 2WD chassis. The HC-SR04 sensor is visible at the front.

C. Arduino Code

The Arduino was programmed to continuously poll the HC-SR04 sensor and update the motor speed, LED state, and buzzer output according to the three-zone logic in Eq. 4. The motor was held off until the push-button was pressed and a valid distance reading ($d > 0$) was returned by the sensor. The key distance thresholds were set at 20 cm and 10 cm, and the motor PWM values were set to 200 (full speed), 120 (slow), and 0 (stop) for the three zones respectively.

V. RESULTS

The robot was tested by driving it toward a wall in a corridor and observing the LED, buzzer, and motor responses. At distances greater than 20 cm the car drove forward at full speed with the green LED illuminated. Upon entering the 10–20 cm zone the car slowed noticeably, the yellow LED activated, and the buzzer emitted an intermittent warning beep. At distances of 10 cm or less, the motors stopped completely, the red LED turned on, and the buzzer sounded continuously.

TABLE I. Observed robot behavior at each distance zone during testing.

Distance	Motor	LED	Buzzer
$d > 20$ cm	Full speed	Green	Off
$10 < d \leq 20$ cm	Slow	Yellow	Intermittent
$d \leq 10$ cm	Stop	Red	Continuous

The system performed reliably at distances of approximately 1–1.5 m from the wall, where the ultrasonic sensor had a clear line of sight to a flat reflective surface. Two notable failure modes were observed during testing. First, in a large open hallway, the sensor occasionally failed to detect any obstacle and returned no valid reading, causing the robot to remain stationary even with the button pressed. Second, at certain angles of approach the sensor would intermittently lose the echo return, causing the car to momentarily accelerate back to full speed before slowing again.

VI. DISCUSSION

The three-zone obstacle avoidance system functioned as designed within its effective operating range. The PWM speed control produced a clear and audible difference between the full-speed and slow-speed states, and the LED and buzzer indicators provided unambiguous feedback at each zone transition, consistent with the logic in Eq. 4.

The two failure modes observed are both attributable to fundamental limitations of the HC-SR04 ultrasonic sensor. The sensor relies on specular reflection of the ultrasonic pulse back toward the receiver. In a large open space, the pulse may spread and dissipate before returning a detectable echo, causing the `pulseIn()` function

to time out and return zero. This explains why the car would not move in the large hallway — no valid distance reading was available, so the motor enable condition was never satisfied. This is actually the safe failure mode: the system correctly refused to drive when sensor data was unavailable.

The intermittent loss of signal at certain approach angles is caused by the narrow effective beam angle of the HC-SR04 ($\approx 15^\circ$). When the robot approached the wall at an oblique angle, the reflected pulse was directed away from the receiver rather than back toward it. This could be mitigated by using multiple sensors at different angles, a wider-beam sensor, or a rotating servo-mounted sensor.

The button logic worked as intended: the motor remained off until the button was pressed, and the system checked for a valid sensor reading before engaging the motors, preventing uncontrolled motion.

VII. CONCLUSION

An autonomous obstacle-avoidance robot car was successfully built and demonstrated using an Arduino Uno, HC-SR04 ultrasonic sensor, and L298N motor driver. The system correctly implemented a three-zone proximity response — full speed, slow with warning, and full stop with alarm — and performed reliably within 1–1.5 m of flat wall surfaces. Sensor limitations inherent to the ultrasonic measurement principle, including beam angle constraints and open-space echo loss, were identified as the primary sources of occasional missed detections. These findings highlight the importance of understanding sensor physics when designing autonomous systems, and suggest that multi-sensor fusion or alternative sensing modalities (e.g., infrared or LiDAR) could improve robustness in future iterations.